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Description - This procedure describes the removal and replacement of the system support module (SSM), the analog processor module (APM), the communications pin driver (CPD), the front panel display, the integrated power supply (IPS), and the rear panel assembly for the 1.5T and 1.0T RF/PDU cabinet.

1 - TOOLS REQUIRED

- #2 Phillips screwdriver
- #1 Phillips screwdriver
- 3/16" nut driver

2 - LOCKOUT / TAGOUT PROCESS

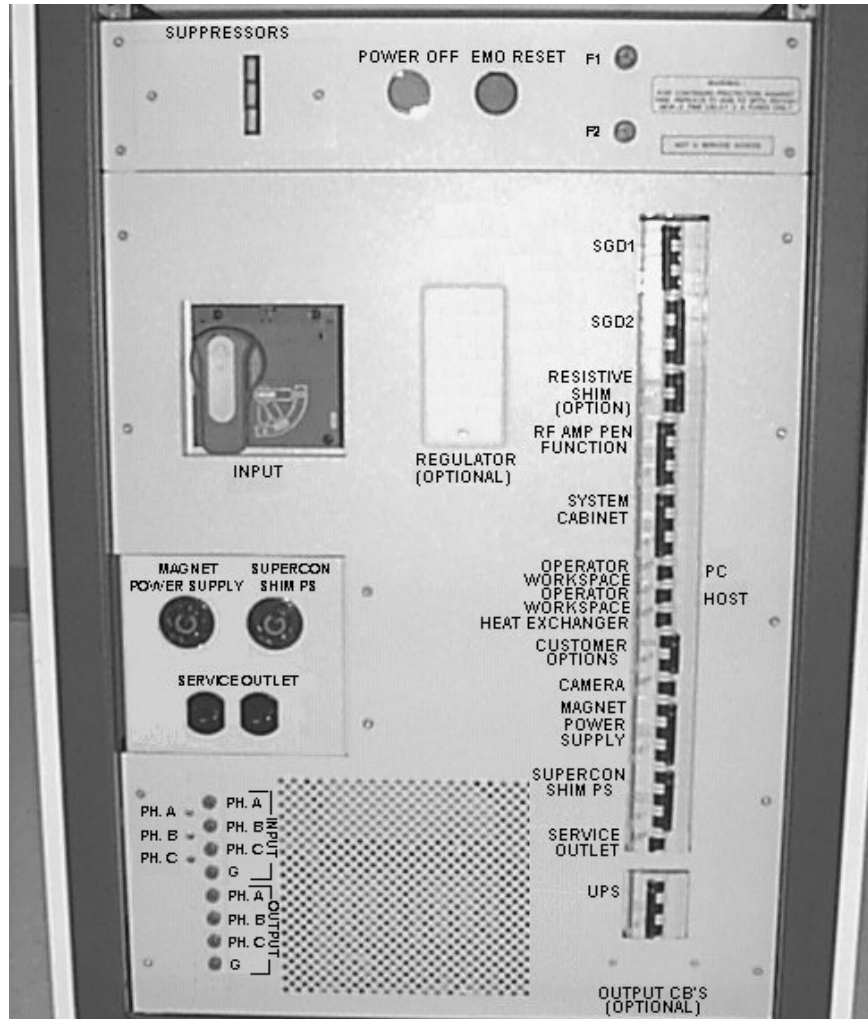


POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE PDU MAIN INPUT CIRCUIT BREAKER BEFORE REMOVING THE SSM. VERIFY THAT LOCK OUT AND TAG OUT OF THE PDU MAIN INPUT CIRCUIT BREAKER IS PROPERLY PERFORMED.

1. Bring the PC software down.
 - a. Toggle mouse control to the PC monitor using the hardkey on the top, left side of the keyboard housing.
 - b. From the PC monitor use the mouse to select **Start → Shutdown → Shut down**.
2. Bring the Signa software down.
 - a. Toggle mouse control back to the Signa monitor using the hardkey on the top, left side of the keyboard.
 - b. Single-click on the **Toolbelt** icon.
 - c. Single-click on the **System Shutdown** softkey on the Service Desktop Manager.
 - d. Select **OK** to confirm the shutdown.
 - e. Wait for the system to indicate on the monitor that it is safe to power off the computer before proceeding. This usually takes about 90 seconds before this message is seen.
3. Remove the cover from the front of the RF/PDU cabinet to expose the front of the PDU.

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

4. Refer to Illustration 2-1 and power off the following circuit breakers:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**
 - e. **OPERATORS WORKSPACE (PC)**
 - f. **OPERATORS WORKSPACE (HOST)**
 - g. **GRADIENT WATER CHILLER**



BREAKER LOCATION
ILLUSTRATION 2-1

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

5. Rotate the larger main breaker labeled INPUT counter-clockwise to the green, O (OFF) position. See Illustration 2-2.



MAIN INPUT BREAKER IN THE OFF POSITION

ILLUSTRATION 2-2

6. Lock Out / Tag Out the PDU "INPUT" circuit breaker.
 - a. Press on the left-pointing arrow on the end of the handle to expose the location where the lock and tag can be placed.

2 - LOCKOUT / TAGOUT PROCESS (CONTINUED)

- b. Lock and tag out the PDU "INPUT" circuit breaker as shown in Illustration 2-3



PDU MAIN INPUT CIRCUIT BREAKER LOCKED AND TAGGED OUT
ILLUSTRATION 2-3

7. Verify all LEDs are OFF (not illuminated) on front of SSM and RFI. Verify all LEDs are OFF (not illuminated) at the rear of the RF Amplifier Power Module.
8. Using an AC voltmeter (DVM), measure the input power leads on the rear of one of the RF Amplifier Modules from the rear (right side) of the RF/PDU Cabinet.
 - a. **If a 1.5T RF/PDU Cabinet:** Measure between L1 and L2; verify voltmeter measures 0 VAC.
 - b. **If a 1.0T RF/PDU Cabinet:** Measure between T1 (black wire) and T2 (red wire); verify voltmeter measures 0 VAC.

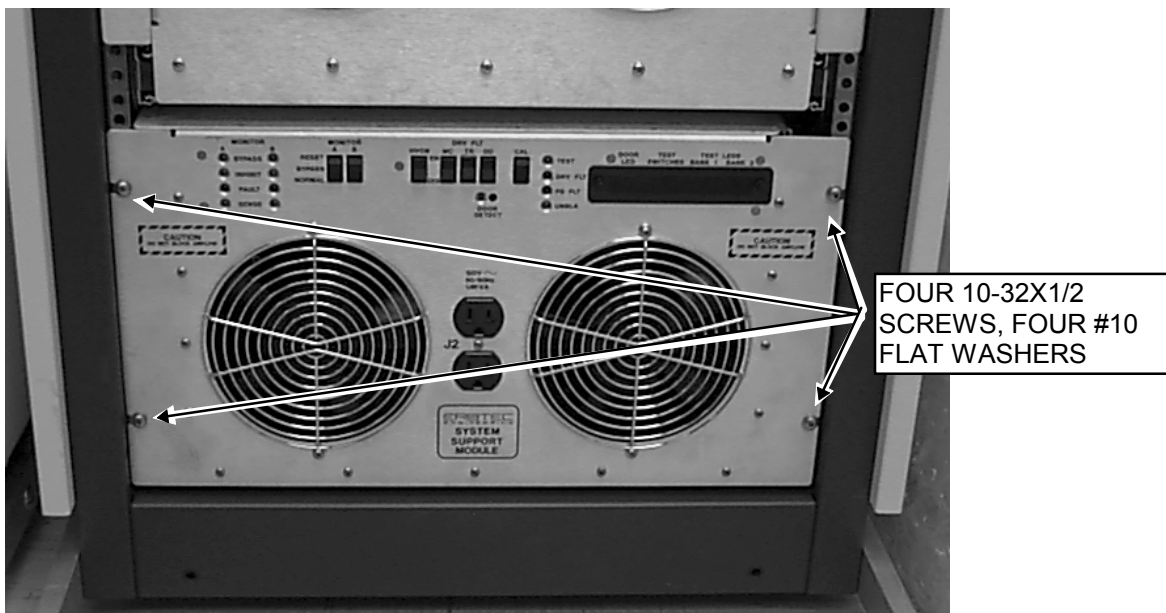
3 – SSM

3-1 Removal of SSM



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE PDU MAIN INPUT CIRCUIT BREAKER BEFORE REMOVING SSM. VERIFY THAT LOCK OUT AND TAG OUT OF THE PDU MAIN INPUT CIRCUIT BREAKER IS PROPERLY PERFORMED.

1. Perform **Section 2 – LOCKOUT / TAGOUT PROCESS** and verify that the cabinet power is off (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Remove the front door from the cabinet.
3. Remove the four 10-32x1/2 screws and four #10 flat washers that secure the SSM chassis to the cabinet (see Illustration 3-1).



**REMOVAL OF SYSTEM SUPPORT MODULE
ILLUSTRATION 3-1**

9. Remove all external cabling from the rear of the SSM.
5. Slide out the SSM from the cabinet.



The SSM weighs about 35 lbs (15.88 kg).

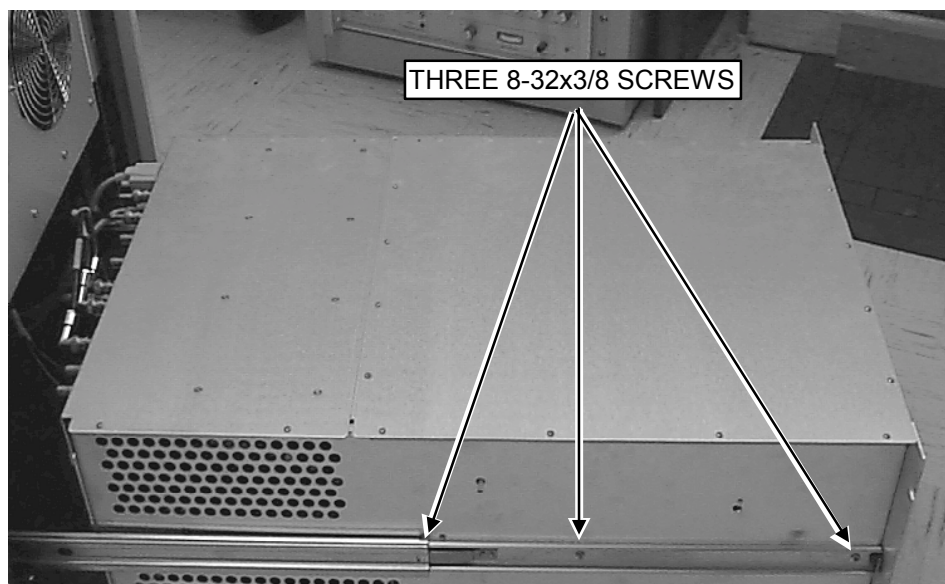
3-1 Removal of SSM (Continued)

6. Use one or more telephone books or blocks of wood to support the module from beneath to alleviate pressure on the slides. The distance from the bottom of the module to the floor is about 6.5 inches (16.5 cm).



DO NOT ALLOW THE MODULE TO DROP TO THE FLOOR.

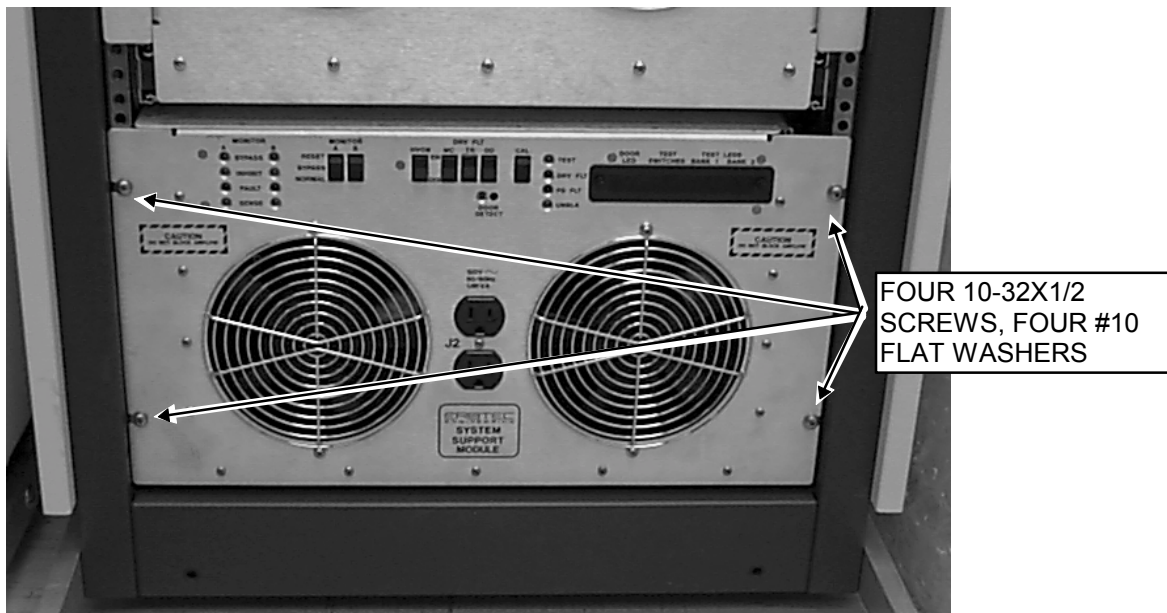
7. Remove the six 8-32x3/8 screws (three per slide) that secure the rack slides to the module. This enables you to completely remove the unit. See Illustration 3-2.



RACK SLIDE SCREW LOCATIONS
ILLUSTRATION 3-2

3-2 Installation of New SSM

1. Support the SSM from beneath to alleviate pressure on the slides.
2. Install the six 8-32x3/8 screws (three per slide) that secure the rack slides to the module.
3. Carefully slide the SSM back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.)
4. Install the four 10-32x1/2 screws and four #10 flat washers that secure the SSM to the cabinet (see Illustration 3-3).



INSTALLATION OF SYSTEM SUPPORT MODULE
ILLUSTRATION 3-3

5. Replace all external cable connections at the rear of the SSM.
6. Verify that all the circuit breakers on the rear of the SSM are in the up (ON) position.
7. Remove Lock Out / Tag Out hardware from the PDU "INPUT" circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.
8. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.
9. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**

3-2 Installation of New SSM (Continued)

- e. **OPERATORS WORKSPACE (PC)**
- f. **OPERATORS WORKSPACE (HOST)**
- g. **GRADIENT WATER CHILLER**

10. Power the Host and PC computers on and then boot the SIGNA and PC software as required.
11. Perform the SSM calibration procedure described in **CPD Board Setup and Calibration** (RC3SCA2.DOC or RC3SCC2.DOC) on the Signa 8.X Service Methods CDROM.
 - a. Perform **Section 2 – PREREQUISITES**.
 - b. Perform **Section 4 – SSM CAL SWITCH**.
 - c. Perform **Section 8 – POWER MONITOR TYPE CONFIGURATION**.
 - d. If necessary, perform **Section 7 – PAC VOLTAGE CONFIGURATION**.
 - e. If site has MN Spectroscopy hardware then refer to **Direction 2194610 Signa Release 8.X Multi-Nuclear Spectroscopy Subsystem**, Setup and Calibrations section, and perform **Sub-section 1-6 MN Spectroscopy TR Driver Adjustment for SSM**. Direction 2194610 can be found on the 8.X Service Methods CD-ROM. Perform the SSM calibration procedure described in **CPD Board Setup and Calibration** (RC3SCA2.DOC or RC3SCC2.DOC) on the Signa 8.X Service Methods CDROM.
12. Perform one head and body scan to verify that the system is operational.
13. Replace rear and front RF/PDU cabinet covers.

4 - REMOVING LID FROM SSM CHASSIS



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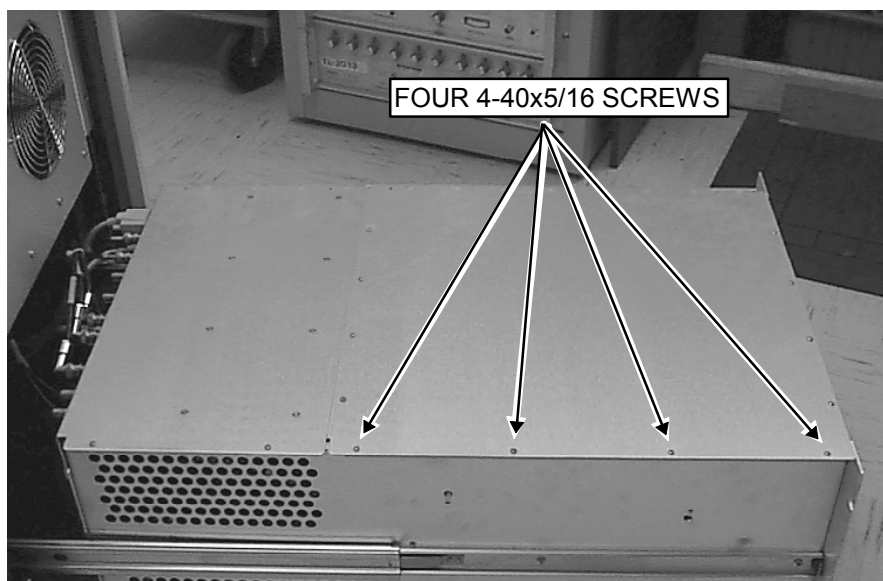
1. Perform **Section 2 - LOCKOUT / TAGOUT PROCESS** and verify that the cabinet power is off (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Slide the SSM fully out of the cabinet as described in **Section 3-1 Removal of SSM** but do not remove the screws or SSM from the slide rails.
3. Remove the four 4-40x5/16 panhead screws that secure the front lid to the front panel (see Illustration 4-1).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM FRONT PANEL
ILLUSTRATION 4-1

4 - REMOVING LID FROM SSM CHASSIS (CONTINUED)

4. Remove the four 4-40x5/16 panhead screws that secure the front lid to the chassis (see Illustration 4-2).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM CHASSIS
ILLUSTRATION 4-2

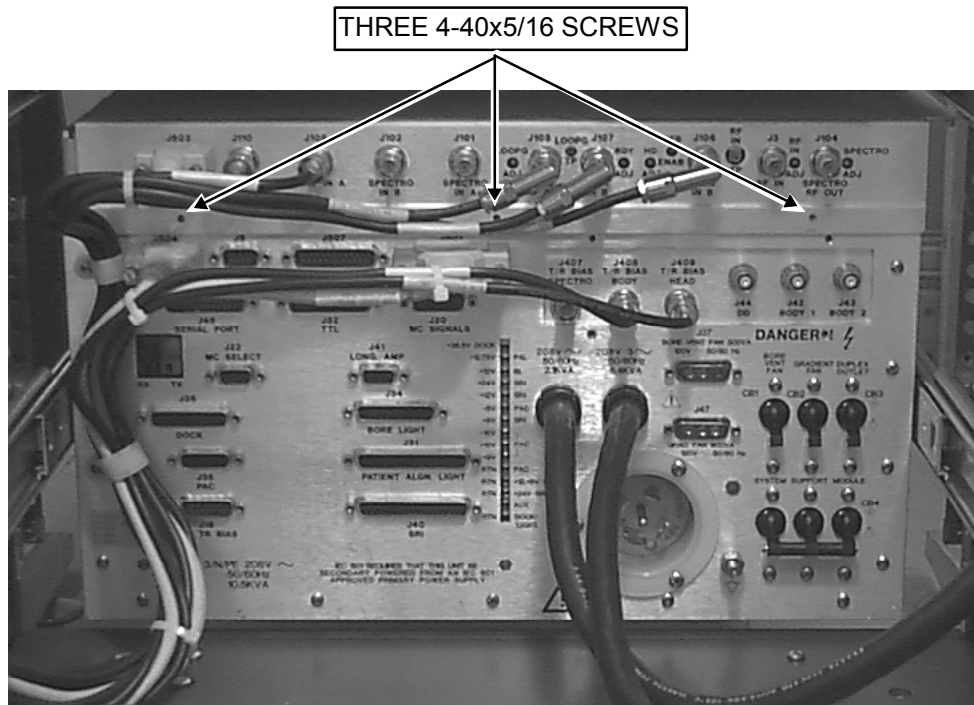
5. Remove the two 4-40x5/16 panhead screws that secure the APM lid to the chassis (see Illustration 4-3).



REMOVAL OF ANALOG PROCESSOR MODULE LID FROM CHASSIS
ILLUSTRATION 4-3

4 - REMOVING LID FROM SSM CHASSIS (CONTINUED)

6. Remove the three 4-40x5/16 panhead screws that secure the APM lid to the rear panel (see Illustration 4-4). A few cables may need to be disconnected first.



REMOVAL OF ANALOG PROCESSOR MODULE LID FROM REAR PANEL
ILLUSTRATION 4-4

7. Grasp the edge of the module lid where the APM lid meets the front lid, and lift the lid to a vertical position. Use care to ensure that the APM cables do not catch on the bottom of the RFI and that there is sufficient slack.

Go to the appropriate section to continue the remove and replace procedure:

See section **5 - Removal of Failed APM Module**

See section **6 - Removal of Failed CPD Module**

See section **7 - Removal of Failed Front Panel Display Board**

See section **8 - Removal of Failed IPS Module**

See section **9 - Removal of Failed Rear Panel Assembly**

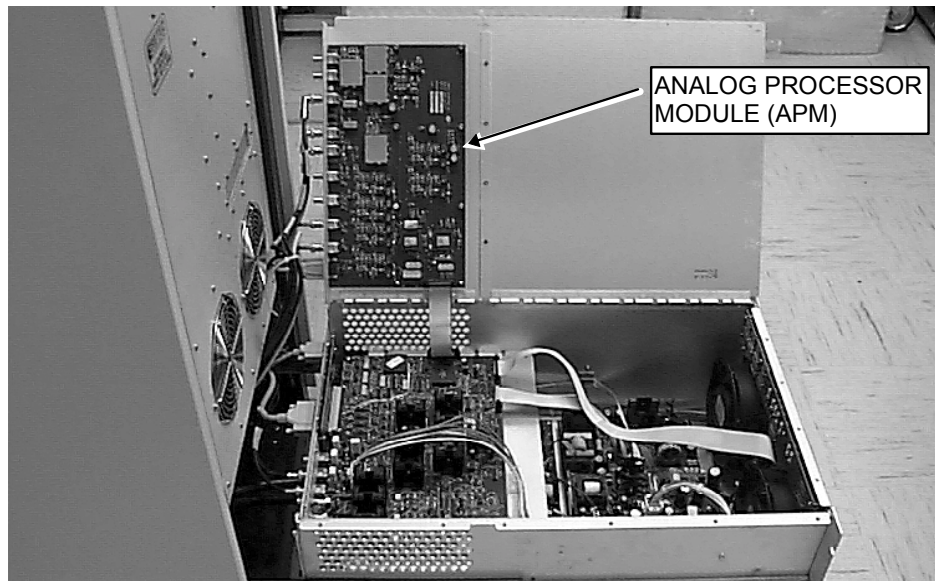
5 - APM MODULE

5-1 Removal of APM Module



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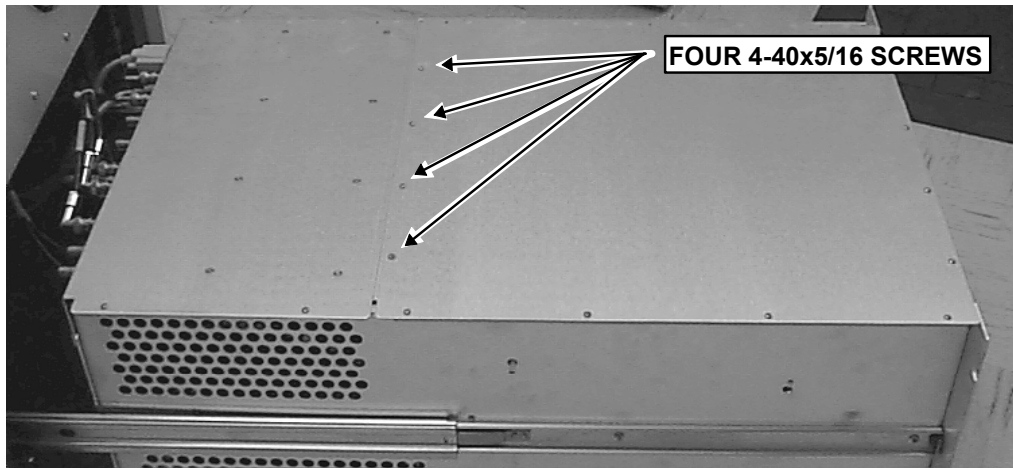
1. Perform **Section 2 - LOCKOUT / TAGOUT PROCESS** and verify that the cabinet power is off (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Remove lid from SSM chassis per instructions in **Section 4 - REMOVING LID FROM SSM CHASSIS**.
3. The APM module is on the lid assembly near the rear of the SSM (see Illustration 5-1).



ANALOG PROCESSOR MODULE LOCATION
ILLUSTRATION 5-1

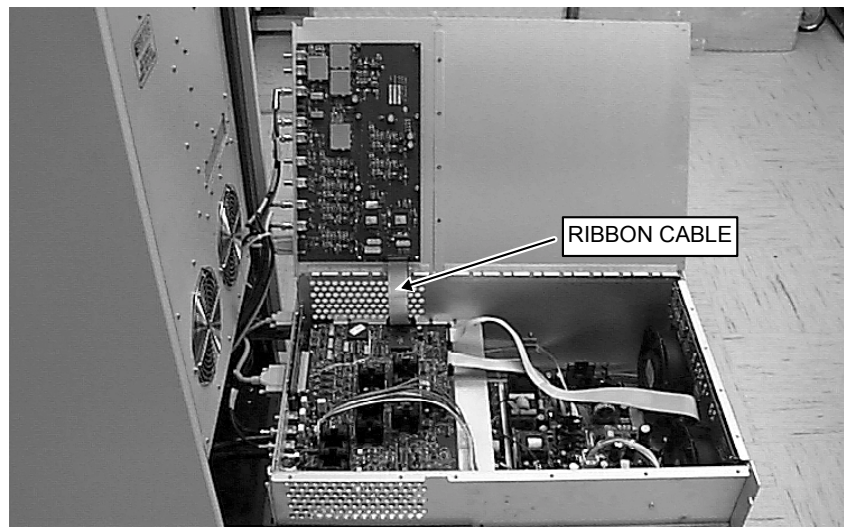
5-1 Removal of APM Module (Continued)

4. Remove the four 4-40 x 5/16 panhead screws that secure the front lid to the APM lid (see Illustration 5-2)



REMOVAL OF ANALOG PROCESSOR MODULE LID FROM FRONT LID
ILLUSTRATION 5-2

5. Disconnect the cables attached to the rear of the APM module and the ribbon cable attached to the APM board. See Illustration 5-3.

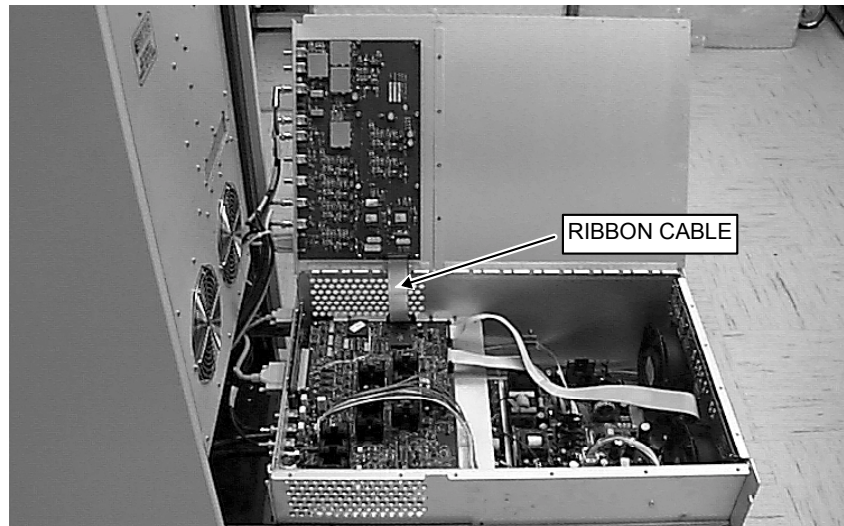


RIBBON CABLE FROM APM BOARD TO CPD BOARD
ILLUSTRATION 5-3

6. Remove the failed APM module by sliding lid forward in the upright position to disengage hinge.

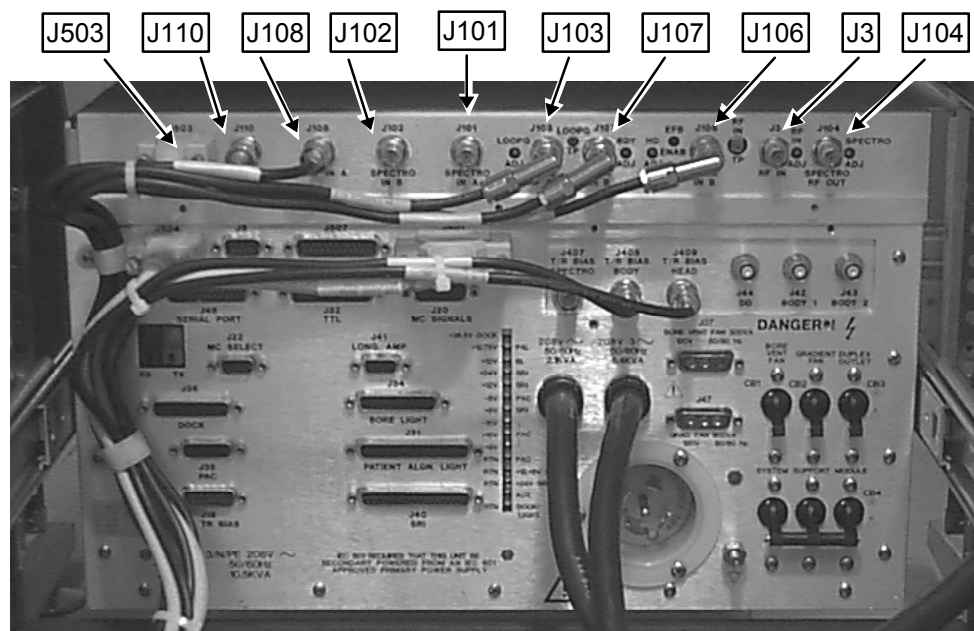
5-2 Installation of APM Module

1. In the upright position, place the APM module on the hinges and slide to the back to attach.
2. Attach the ribbon cable from the CPD board to the APM board. Close APM lid. See Illustration 5-4.



RIBBON CABLE FROM APM BOARD TO CPD BOARD
ILLUSTRATION 5-4

3. Attach the cables to the APM board (see Illustration 5-5). Be sure to match the label designators on the cables to the APM connectors.



ANALOG PROCESSOR MODULE CABLE CONNECTIONS
ILLUSTRATION 5-5

5-2 Installation of APM Module (Continued)

4. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
5. Close the front lid. Install eight 4-40 x 5/16 panhead screws: four to the APM lid, and four to the front panel.
6. Attach six 4-40x5/16 panhead screws that secure both lids to chassis. Tighten all screws in steps 4, 5, and 6.
7. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
8. Verify that all the circuit breakers on the rear of the SSM are in the up (ON) position.
9. Remove Lock Out / Tag Out hardware from the PDU "INPUT" circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.
10. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.
11. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**
 - e. **OPERATORS WORKSPACE (PC)**
 - f. **OPERATORS WORKSPACE (HOST)**
 - g. **GRADIENT WATER CHILLER**
12. Power the Host and PC computers on and then boot the SIGNA and PC software as required.
13. Perform the SSM calibration procedure described in **CPD Board Setup and Calibration** (RC3SCA2.DOC or RC3SCC2.DOC) on the Signa 8.X Service Methods CDROM.
14. Perform one head and body scan to verify that the system is operational.
15. Replace front and rear covers.

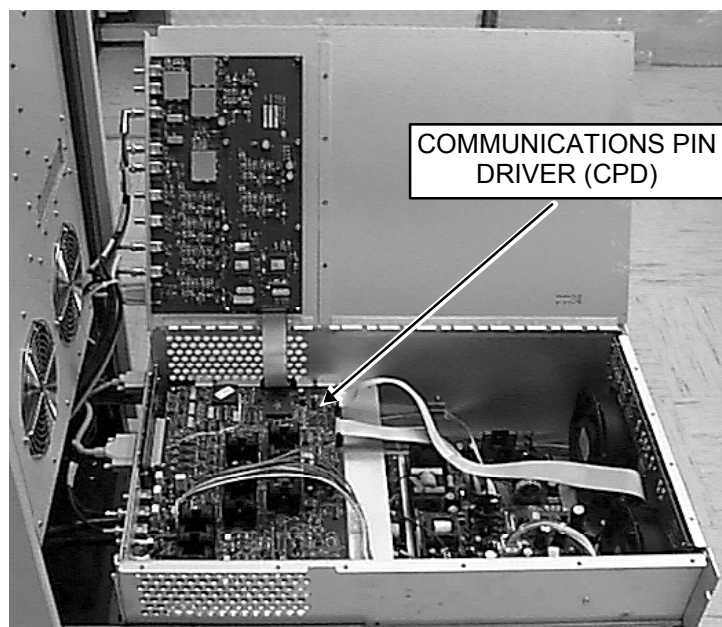
6 - CPD MODULE

6-1 Removal of CPD Module



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1. Perform **Section 2 - LOCKOUT / TAGOUT PROCESS** and verify that the cabinet power is off (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Remove lid from SSM chassis per instructions in **Section 4 - REMOVING LID FROM SSM CHASSIS**.
3. The CPD module is located at the rear of the SSM (see Illustration 6-1).



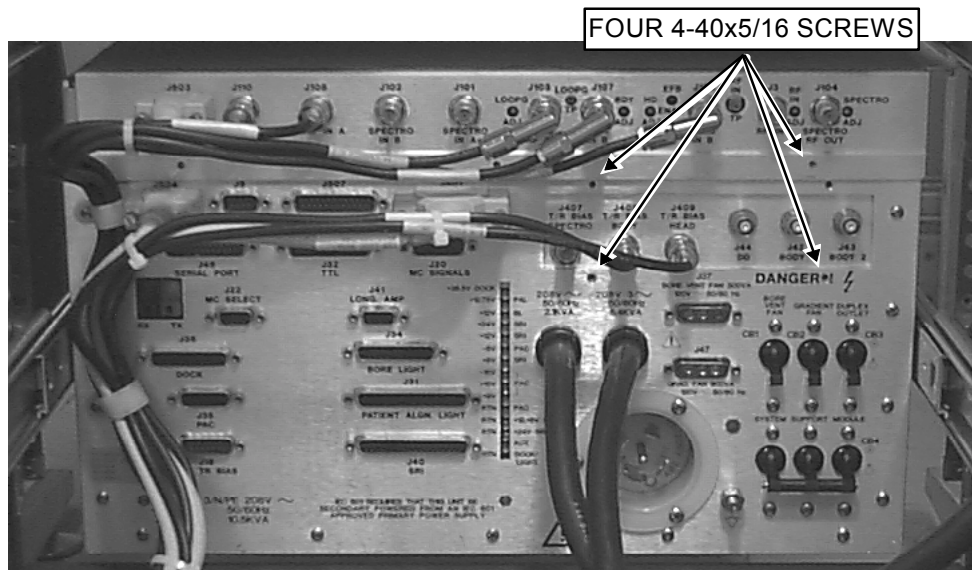
COMMUNICATIONS PIN DRIVER LOCATION

ILLUSTRATION 6-1

4. Disconnect all connectors attached to the CPD board and to the rear of the CPD module.

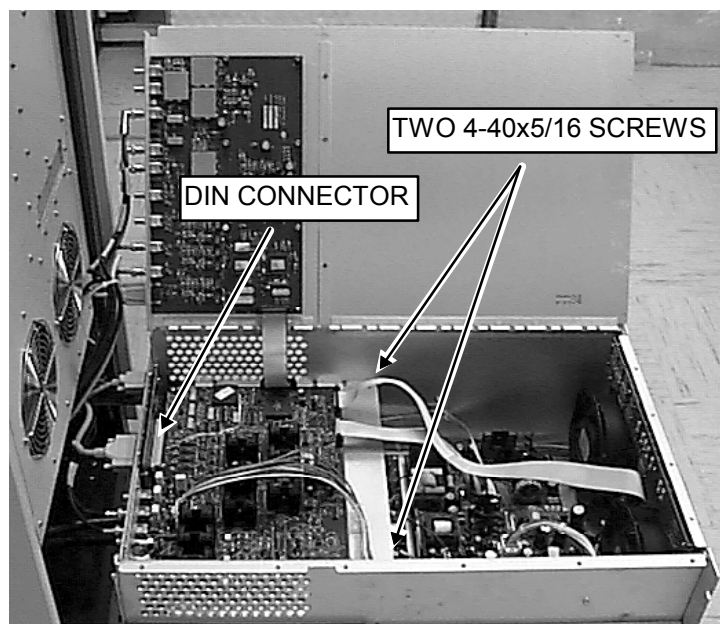
6-1 Removal of CPD Module (Continued)

5. Remove four 4-40x5/16 panhead screws that secure the CPD module to the rear panel (see Illustration 6-2).



REMOVAL OF COMMUNICATIONS PIN DRIVER FROM REAR PANEL
ILLUSTRATION 6-2

6. Remove the two 4-40x5/16 panhead screws that secure the module to the side brackets (see Illustration 6-3).

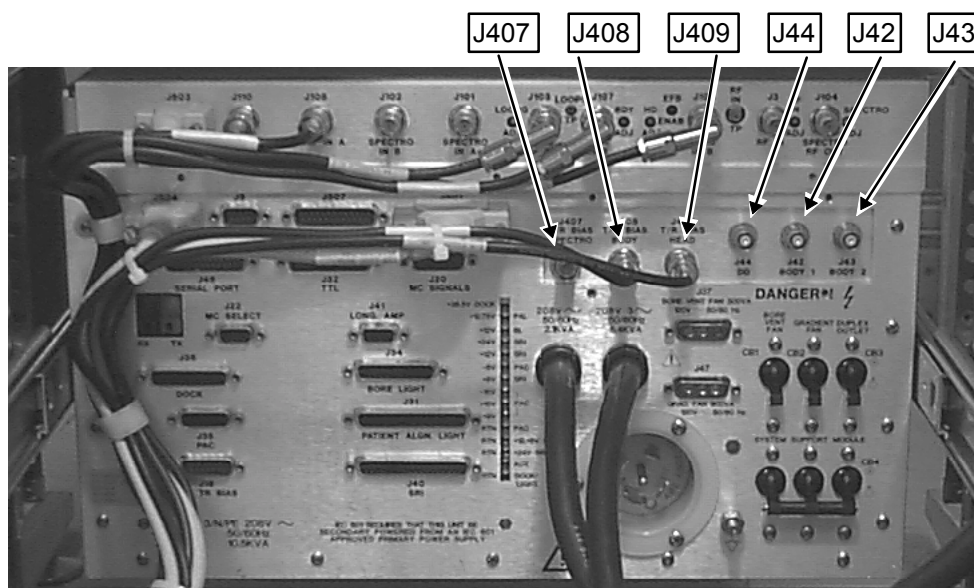


REMOVAL OF COMMUNICATIONS PIN DRIVER FROM SIDE BRACKETS
ILLUSTRATION 6-3

7. Slide CPD module toward the front of the SSM, carefully detaching DIN connector; lift and remove failed CPD module.

6-2 Installation of CPD Module

1. Place the CPD module on the brackets, aligning the slots, and slide toward the rear panel, carefully attaching the DIN connector to the rear panel.
2. Install four 4-40x5/16 screws through the outside of the rear panel into the CPD module. Do not fully tighten screws..
3. Install two 4-40x5/16 panhead screws that secure the module to the side brackets. Tighten all screws in steps 2 and 3.
4. Attach all connections to the CPD board and to the rear of the CPD module (see Illustration 6-4).



COMMUNICATIONS PIN DRIVER CABLE CONNECTIONS
ILLUSTRATION 6-4

5. Close the module lids.
6. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel. Do not fully tighten screws.
7. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel. Do not fully tighten screws.
8. Install six 4-40x5/16 panhead screws that secure both lids to the chassis. Tighten all screws in steps 6, 7, and 8.
9. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
10. Verify that all the circuit breakers on the rear of the SSM are in the up (ON) position.

6-2 Installation of CPD Module (Continued)

11. Remove Lock Out / Tag Out hardware from the PDU "INPUT" circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.
12. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.
13. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**
 - e. **OPERATORS WORKSPACE (PC)**
 - f. **OPERATORS WORKSPACE (HOST)**
 - g. **GRADIENT WATER CHILLER**
14. Power the Host and PC computers on and then boot the SIGNA and PC software as required.
15. Perform the SSM calibration procedure described in **CPD Board Setup and Calibration** (RC3SCA2.DOC or RC3SCC2.DOC) on the Signa 8.X Service Methods CDROM.
 - a. Perform **Section 2 – PREREQUISITES**.
 - b. Perform **Section 4 – SSM CAL SWITCH**.
 - c. Perform **Section 8 – POWER MONITOR TYPE CONFIGURATION**.
 - d. If necessary, perform **Section 7 – PAC VOLTAGE CONFIGURATION**.
 - e. If site has MN Spectroscopy hardware then refer to **Direction 2194610 Signa Release 8.X Multi-Nuclear Spectroscopy Subsystem**, Setup and Calibrations section, and perform **Sub-section 1-6 MN Spectroscopy TR Driver Adjustment for SSM**. Direction 2194610 can be found on the 8.X Service Methods CD-ROM.
16. Perform one head and body scan to verify that the system is operational.
17. Replace the front and rear covers.

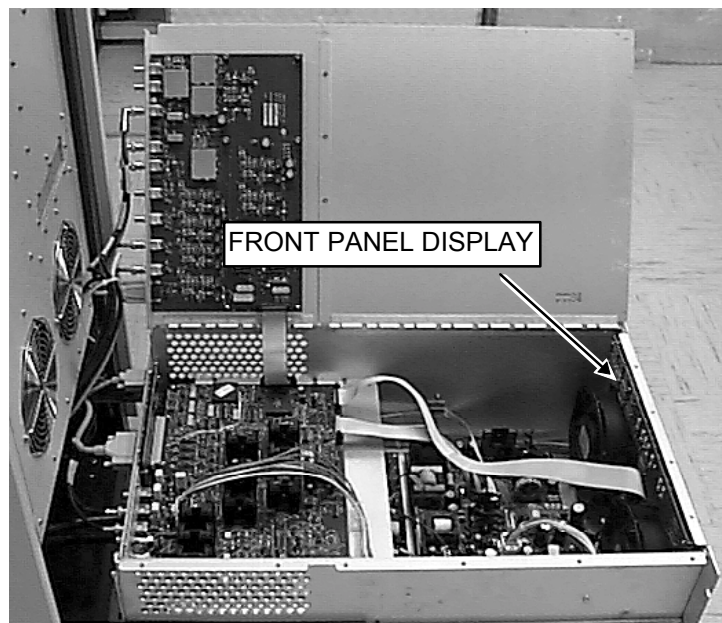
7 - FRONT PANEL DISPLAY BOARD

7-1 Removal of Front Panel Display Board



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1. Perform **Section 2 - LOCKOUT / TAGOUT PROCESS** and verify that the cabinet power is off (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Remove lid from SSM chassis per instructions in **Section 4 - REMOVING LID FROM SSM CHASSIS**.
3. The front panel display board is located on the front panel (see Illustration 7-1).

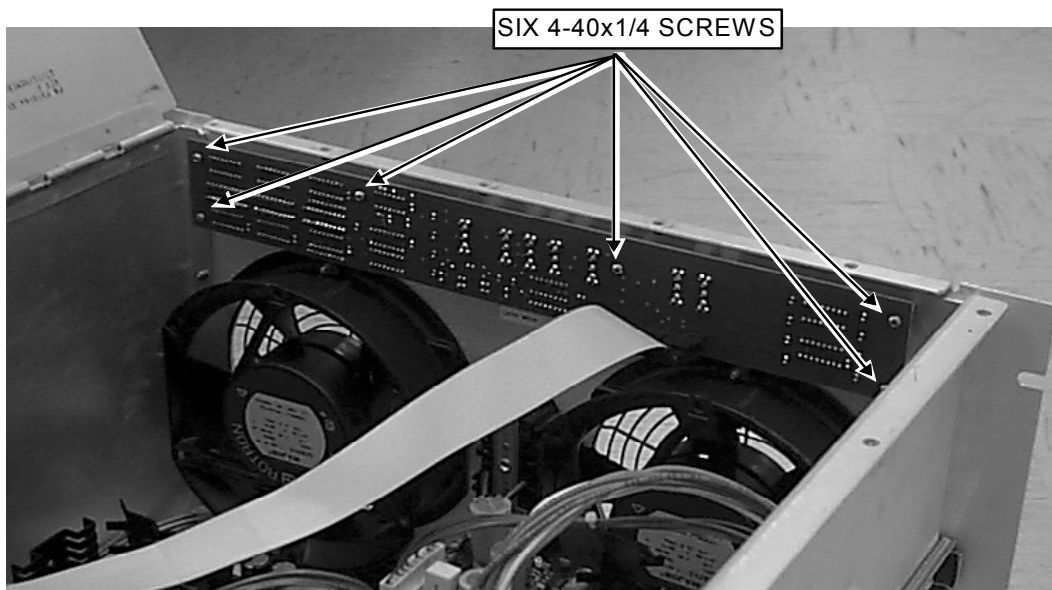


FRONT PANEL DISPLAY LOCATION
ILLUSTRATION 7-1

4. Disconnect the connector attached to the front panel display board.

7-1 Removal of Front Panel Display Board (Continued)

5. Remove six 4-40x1/4 panhead screws that attach the front panel display board to front panel (see Illustration 7-2).



**REMOVAL OF FRONT PANEL DISPLAY
ILLUSTRATION 7-2**

6. Remove the failed front panel display board.

7-2 Installation of Front Panel Display Board

1. Install the front panel display board on front panel.
2. Install and tighten six 4-40x1/4 panhead screws that secure the board to the front panel.
3. Attach the connector from the CPD board to the front panel display board. Close lids.
4. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
5. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel.
6. Install six 4-40x5/16 panhead screws that secure both lids to chassis. Tighten all screws in steps 4, 5, and 6.
7. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
8. Verify that all the circuit breakers on the rear of the SSM are in the up (ON) position.
9. Remove Lock Out / Tag Out hardware from the PDU "INPUT" circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.
10. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.
11. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**
 - e. **OPERATORS WORKSPACE (PC)**
 - f. **OPERATORS WORKSPACE (HOST)**
 - g. **GRADIENT WATER CHILLER**
12. Power the Host and PC computers on and then boot the SIGNA and PC software as required.
13. Perform one head and body scan to verify that the system is operational.
14. Replace front and rear covers.

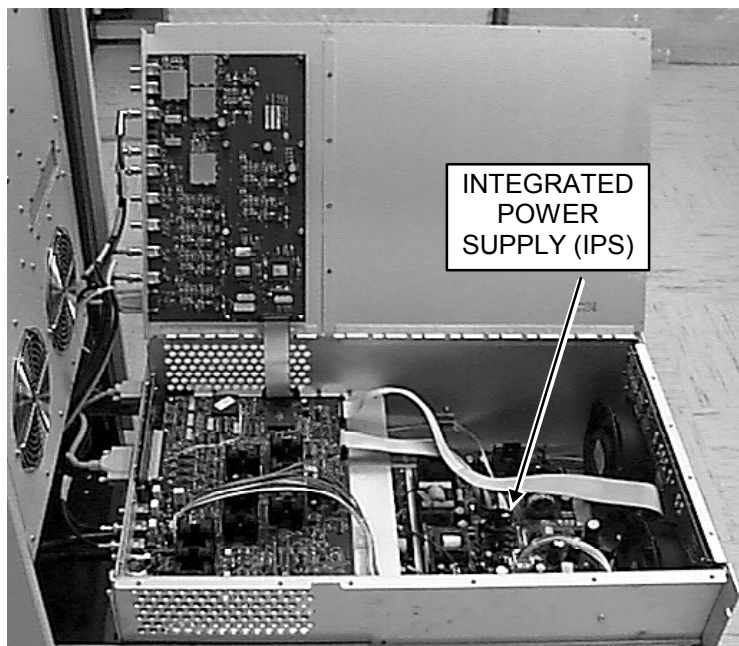
8 - IPS MODULE

8-1 Removal of IPS Module



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1. Perform **Section 2 - LOCKOUT / TAGOUT PROCESS** and verify that the cabinet power is off (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Remove lid from SSM chassis per instructions in **Section 4 - REMOVING LID FROM SSM CHASSIS**.
3. The IPS module is located on the bottom of the SSM (see Illustration 8-1).

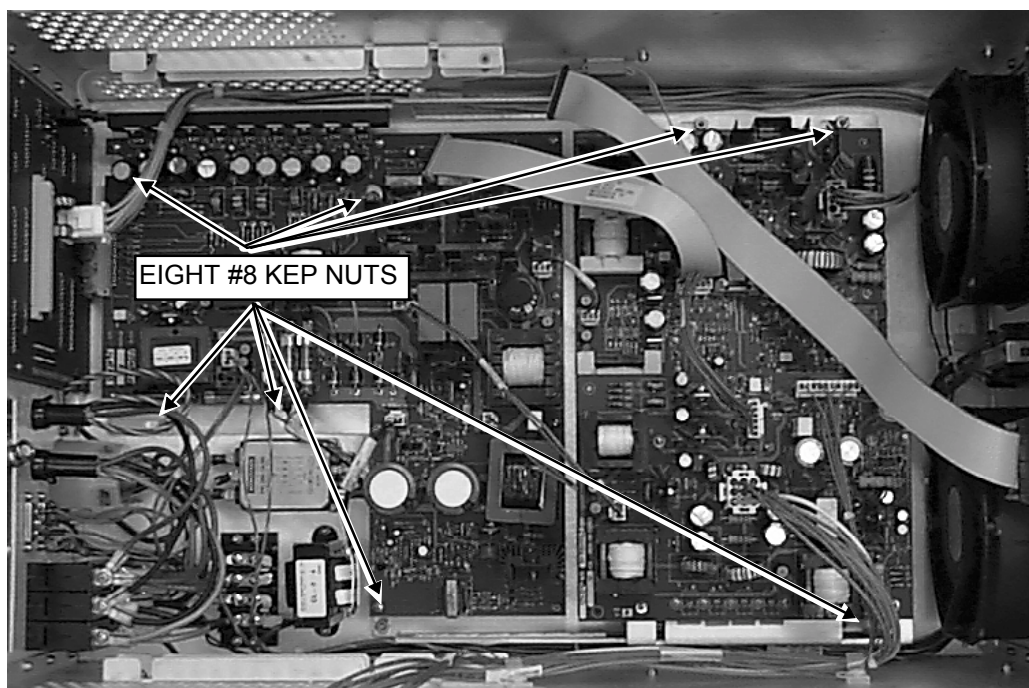


INTEGRATED POWER SUPPLY LOCATION
ILLUSTRATION 8-1

4. The CPD Board will have to be removed before the IPS can be removed. See **Section 6-1 Removal of CPD Module** for instructions on removing the CPD.
5. Disconnect all cables attached to the IPS boards, except those that are internal to the IPS assembly.

8-1 Removal of IPS Module (Continued)

6. Using a nut driver, remove the eight #8 KEP nuts that secure the IPS bulkhead to the bottom of the chassis (see Illustration 8-2).



REMOVAL OF INTEGRATED POWER SUPPLY
ILLUSTRATION 8-2

7. Gently pull the tray up and toward the front in order to disengage the DIN connector on the rear panel while clearing the alignment pins.
8. Lift the entire IPS module out of the chassis.

8-2 Installation of IPS Module

1. Make sure the CPD module has been removed before trying to replace the IPS module.
2. Carefully lower the IPS module into the chassis, aligning the bulkhead slots with the chassis standoffs, and engaging the DIN connector on the rear panel.
3. Secure the module to the chassis with eight #8 KEP nuts.
4. Connect all cabling to the IPS module.

Note

For 1.0T systems ONLY, the JP1 jumper on the HV Board needs to be set to 500 Volts.

5. Replace the CPD module as in **Section 6-2 Installation of CPD Module**.
6. Close the module lids.
7. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel. Do not fully tighten screws.
8. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel. Do not fully tighten screws.
9. Install six 4-40x5/16 panhead screws that secure both lids to the chassis. Tighten all screws in steps 7, 8, and 9.
10. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
11. Verify that all the circuit breakers on the rear of the SSM are in the up (ON) position.
12. Remove Lock Out / Tag Out hardware from the PDU "INPUT" circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.
13. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.
14. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**
 - e. **OPERATORS WORKSPACE (PC)**

8-2 Installation of IPS Module (Continued)

f. OPERATORS WORKSPACE (HOST)

g. GRADIENT WATER CHILLER

15. Power the Host and PC computers on and then boot the SIGNA and PC software as required.
16. Perform **Section 4 – SSM CAL SWITCH** described in **CPD Board Setup and Calibration** (RC3SCA2.DOC or RC3SCC2.DOC) on the Signa 8.X Service Methods CDROM.
17. Perform one head and body scan to verify that the system is operational.
18. Replace front and rear covers.

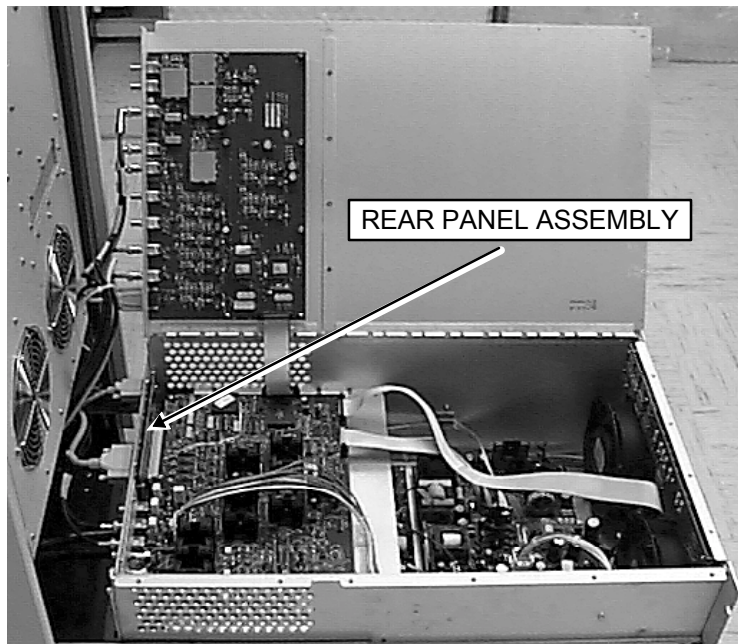
9 - REAR PANEL ASSEMBLY

9-1 Removal of Rear Panel Assembly



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE PDU MAIN INPUT CIRCUIT BREAKER BEFORE WORKING ON THE SSM. VERIFY THAT LOCK OUT AND TAG OUT OF THE PDU MAIN INPUT CIRCUIT BREAKER IS PROPERLY PERFORMED.

1. Perform **Section 2 - LOCKOUT / TAGOUT PROCESS** and verify that the cabinet power is off (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Remove lid from SSM chassis per instructions in **Section 4 - REMOVING LID FROM SSM CHASSIS**.
3. The rear panel assembly is located on the rear of the SSM (see Illustration 9-1).

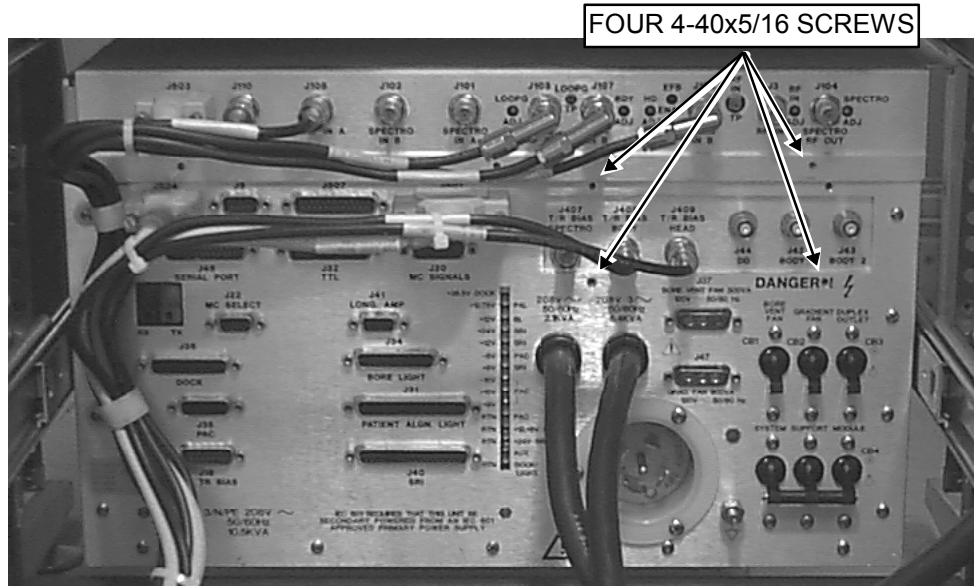


REAR PANEL ASSEMBLY LOCATION
ILLUSTRATION 9-1

4. Disconnect all cables on the outside of the rear panel, except for those connected to the APM.

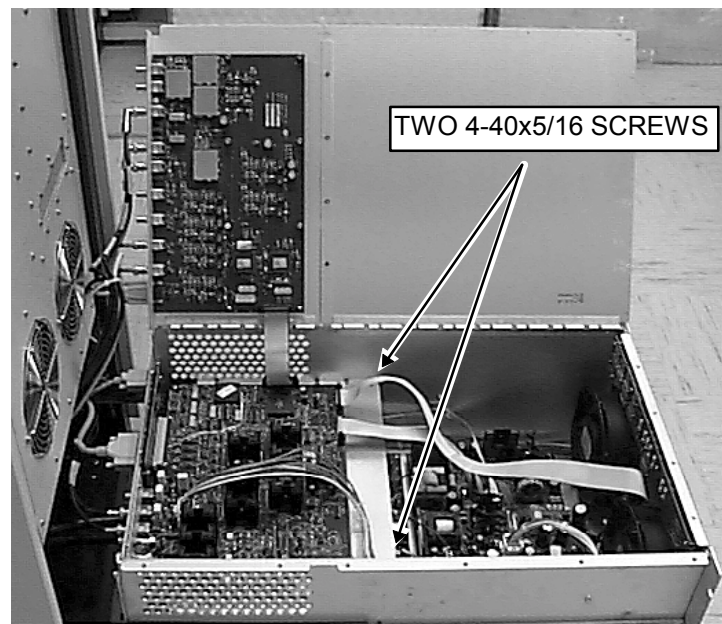
9-1 Removal of Rear Panel Assembly (Continued)

5. Remove four 4-40x5/16 panhead screws that secure the CPD module to the rear panel (see Illustration 9-2).



REMOVAL OF COMMUNICATIONS PIN DRIVER FROM REAR PANEL
ILLUSTRATION 9-2

6. Remove the two 4-40x5/16 panhead screws that secure the CPD module to the side brackets (see Illustration 9-3).

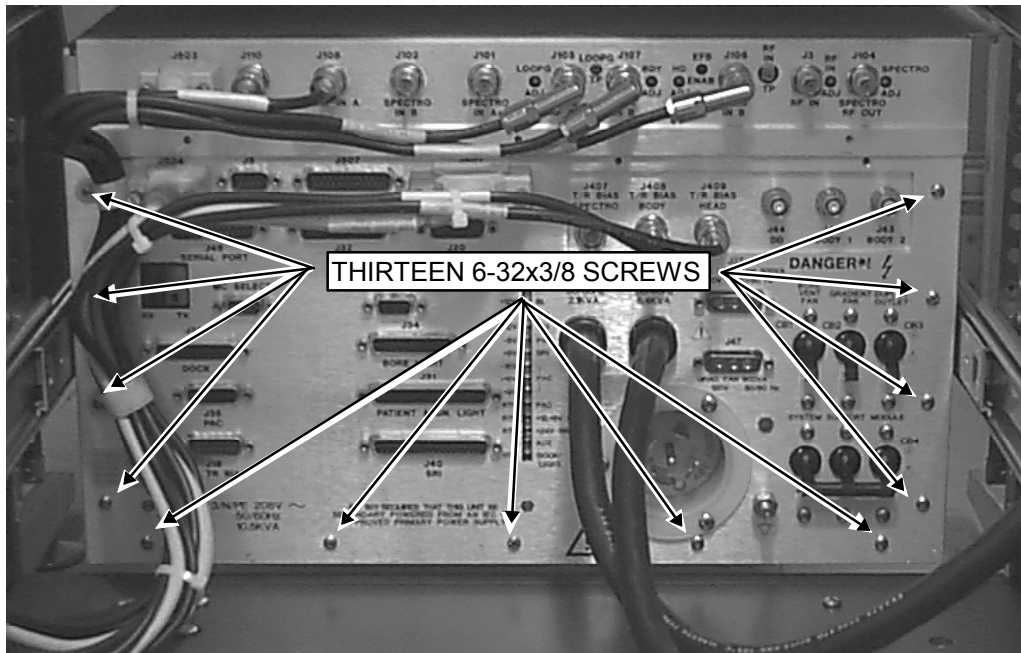


REMOVAL OF COMMUNICATIONS PIN DRIVER FROM SIDE BRACKETS
ILLUSTRATION 9-3

7. Slide the CPD module toward the front of the SSM, carefully detaching the DIN connector.

9-1 Removal of Rear Panel Assembly (Continued)

8. Disconnect all internal cables to the rear interface board, including the DIN connector on the IPS module.
9. Remove thirteen 6-32x3/8 panhead screws that secure the rear panel to the chassis (see Illustration 9-4).



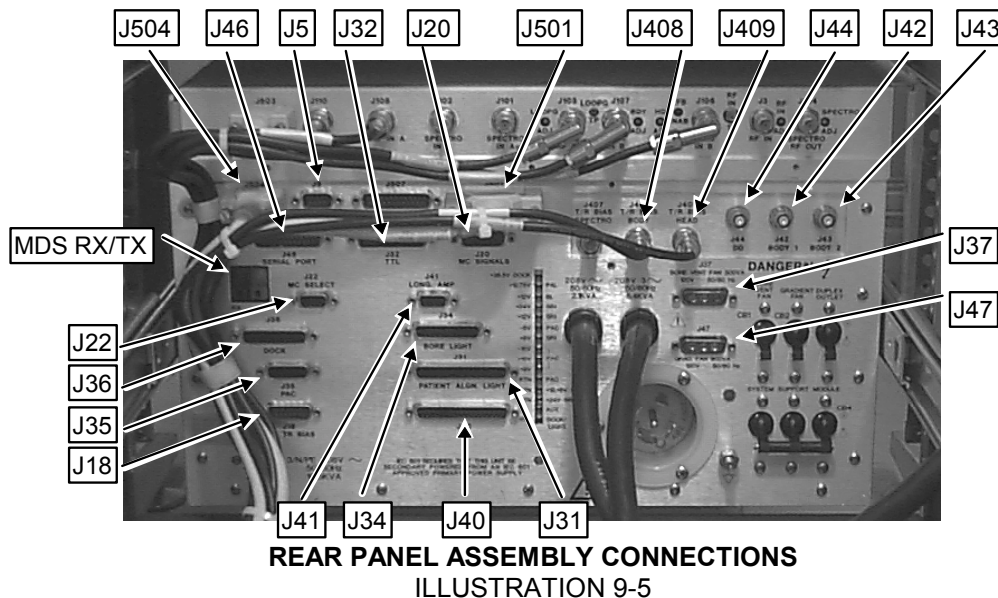
REMOVAL OF REAR PANEL ASSEMBLY

ILLUSTRATION 9-4

10. Remove one 4-40x5/16 panhead screw on the front panel that secures the duplex connector, and disconnect power to the fans. Release the harness from all cable clamps.
11. Remove the failed rear panel assembly and the associated cable harnesses.

9-2 Installation of Rear Panel Assembly

1. Install thirteen 6-32x3/8 panhead screws that secure the rear panel to the chassis.
2. Install one 4-40x5/16 panhead screw on the front panel that secures the duplex connector, and connect power to the fans. Attach all cable clamps to the harness.
3. Connect all internal cables to the rear interface board, including the DIN connector on the IPS module.
4. Slide the CPD board toward the rear panel, and plug in the DIN connector. Attach four 4-40x5/16 panhead screws that secure the CPD board to the rear panel. Attach two 4-40x5/16 panhead screws that secure the CPD module to the side brackets.
5. Attach all external connections (see Illustration 9-5).



6. Close the module lids.
7. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel. Do not fully tighten screws.
8. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel. Do not fully tighten screws.
9. Install six 4-40x5/16 panhead screws that secure both lids to the chassis. Tighten all screws in steps 6, 7, and 8.
10. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
11. Verify that all the circuit breakers on the rear of the SSM are in the up (ON) position.

9-2 Installation of Rear Panel Assembly (Continued)

12. Remove Lock Out / Tag Out hardware from the PDU “INPUT” circuit breaker and rotate the breaker fully clockwise into the red, 1 (ON) position to restore power to the system.
13. Push the red **EMO RESET** button on the top, front of the PDU once to reset the emergency stop circuit.
14. Place the following circuit breakers on the front of the Power Distribution Unit (PDU) into the right (ON) position:
 - a. **SGD 1**
 - b. **SGD 2**
 - c. **RFI / AMP** (shown as RF Amp Pen Function in Illustration 2-1)
 - d. **SYSTEM SUPPORT MODULE**
 - e. **OPERATORS WORKSPACE (PC)**
 - f. **OPERATORS WORKSPACE (HOST)**
 - g. **GRADIENT WATER CHILLER**
15. Power the Host and PC computers on and then boot the SIGNA and PC software as required.
16. Perform one head and body scan to verify that the system is operational.
17. Replace front and rear covers.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	July 23, 1997 Feb 27, 1998	Verdon K.Keshena	Initial version. Updated to RF/PDU Cabinet
B	May 13, 1998	wek	Updated lock out/tag out info
0	June 30, 1998	Erbtec Engineering	Initial release. (KK)
1	January 26, 1999	K. Keshena	Added Note for jumper JP1 setting for the High Voltage Board in Section 13- Installation of New IPS Module.
2	January 4, 2001	D. Thome	Changed Lockout/Tagout section.
3	May 3, 2001	D. Thome	Added more detail to Lockout/Tagout and calibration instructions. Corrected errors.
4	May 15, 2001	D. Thome	Modified Lockout/Tagout information.
5	Aug. 15, 2001	D. Thome	Added T1, T2 power connections to Lockout/Tagout.